

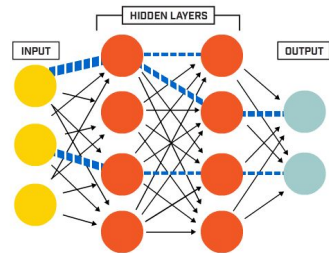


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LSTM



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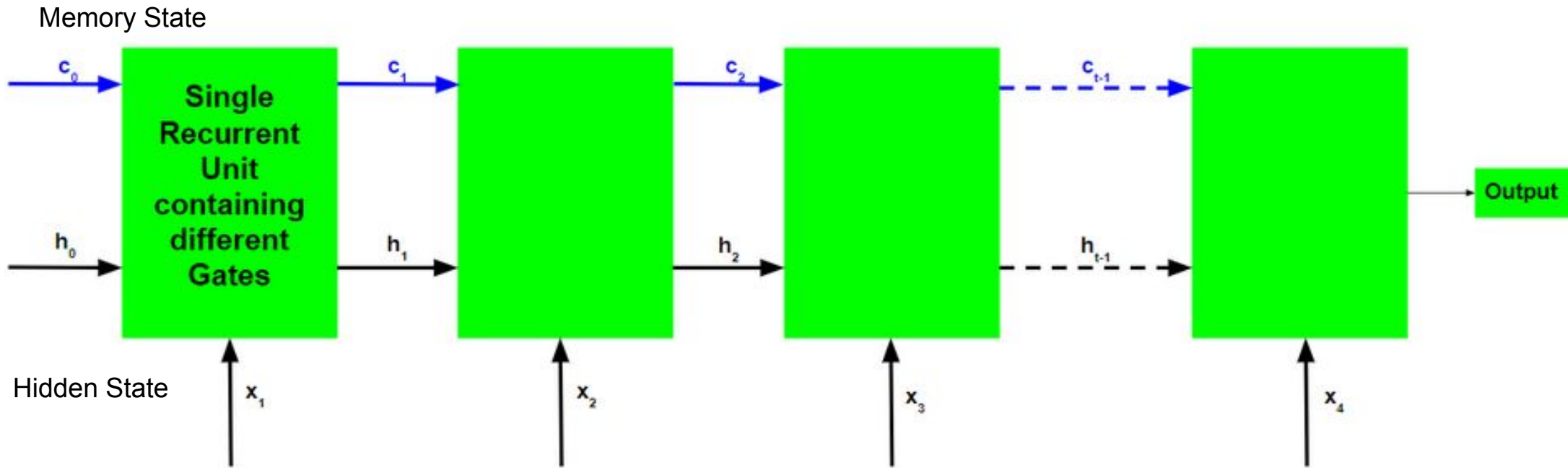


LSTM – Long Short Term Memory

- Vanishing Gradient Problem
- LSTMs Long Short-Term Memory is a type of RNNs Recurrent Neural Network that can detain long-term dependencies in sequential data
- An LSTM recurrent unit tries to “remember” all the past knowledge that the network is seen so far and to “forget” irrelevant data.
- This is done by introducing different activation function layers called “gates” for different purposes.
- Each LSTM recurrent unit also maintains a vector called the Internal Cell State which conceptually describes the information that was chosen to be retained by the previous LSTM recurrent unit.



LSTM – Long Short Term Memory





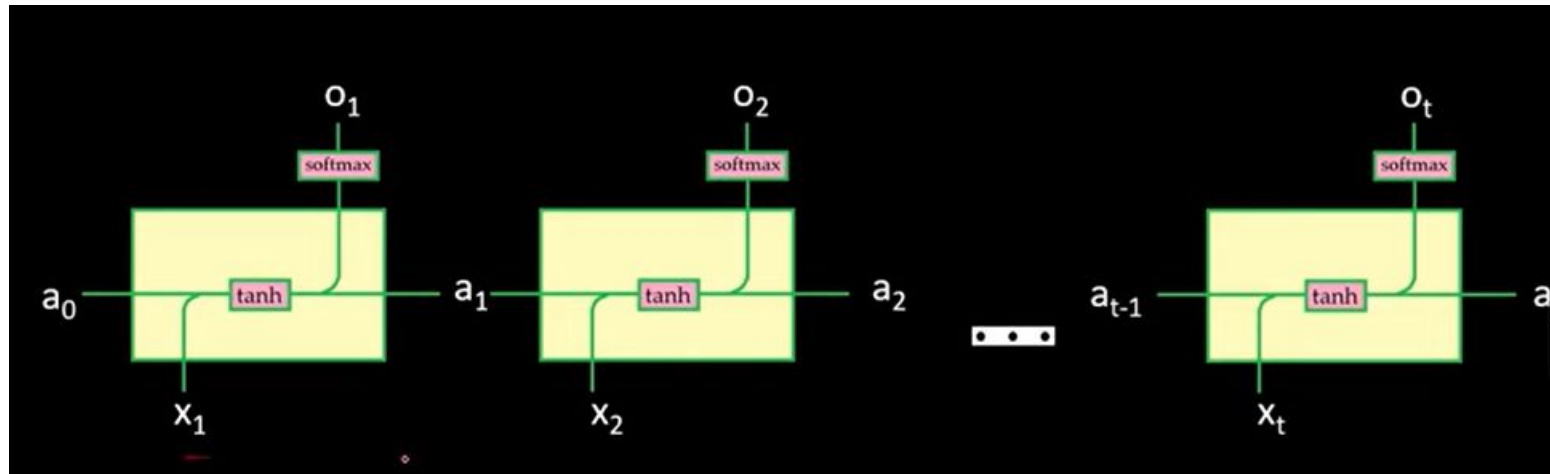
LSTM – Long Short Term Memory

1. **Forget Gate(f):** At forget gate the input is combined with the previous output to generate a fraction between 0 and 1, that determines how much of the previous state need to be preserved (or in other words, how much of the state should be forgotten). This output is then multiplied with the previous state.
2. **Input Gate(i):** Input gate operates on the same signals as the forget gate, but here the objective is to decide which new information is going to enter the state of LSTM. The output of the input gate (again a fraction between 0 and 1) is multiplied with the output of tan h block that produces the new values that must be added to previous state. This gated vector is then added to previous state to generate current state
3. **Output Gate(o):** At output gate, the input and previous state are gated as before to generate another scaling fraction that is combined with the output of tanh block that brings the current state. This output is then given out. The output and state are fed back into the LSTM block.

LSTM – Long Short Term Memory

- Consider a grammar Checker

Mr X goes to play cricket every day and her daughter is also interested

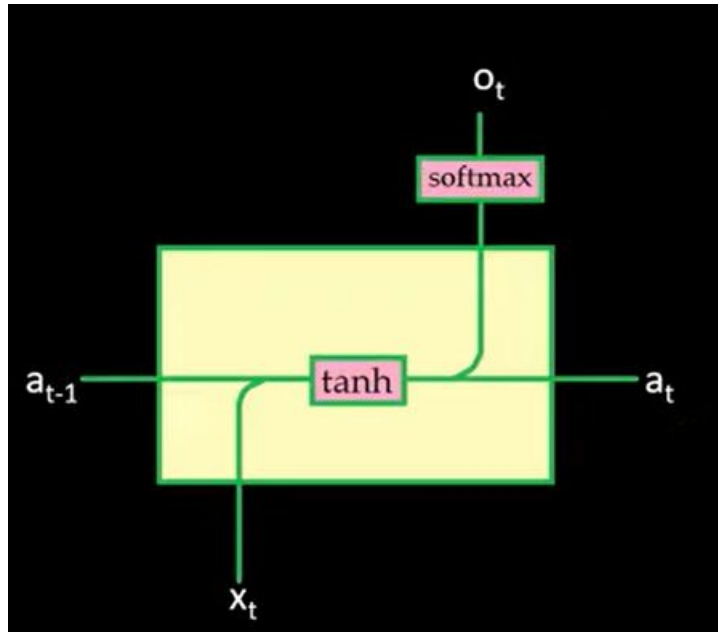


A Typical RNN

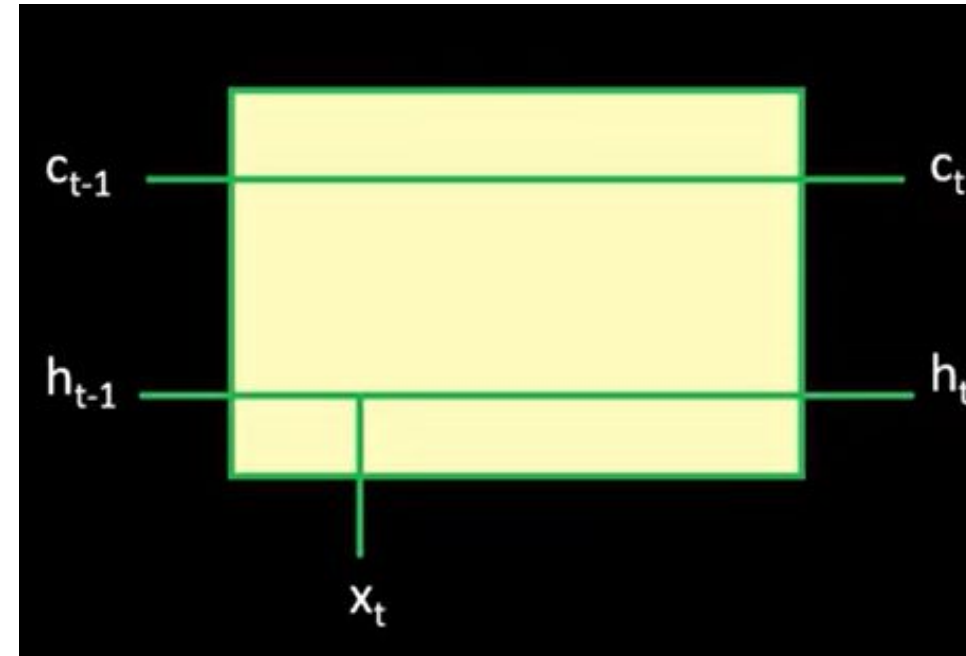
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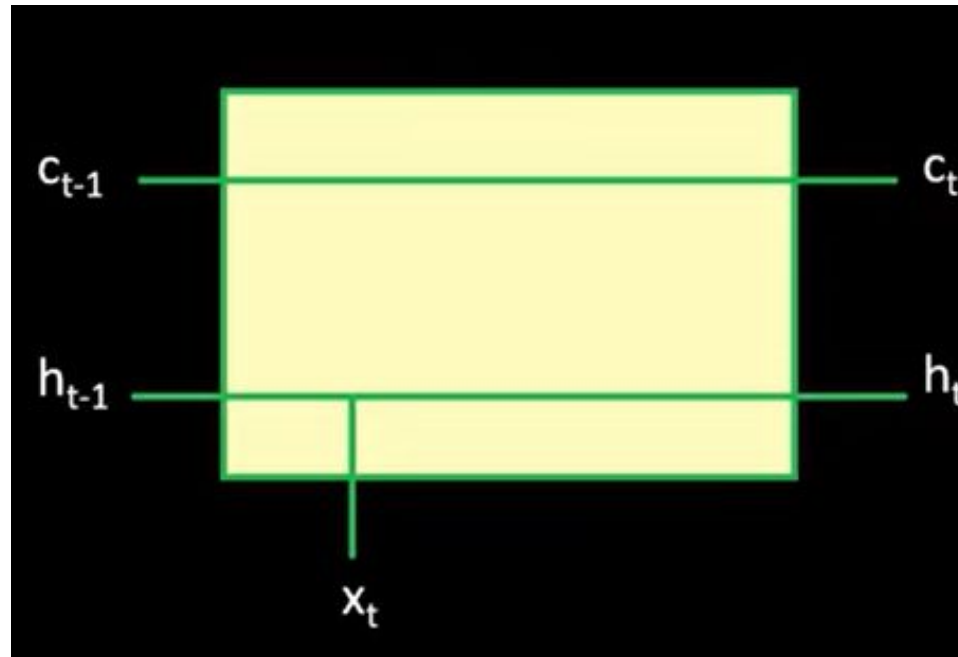


LSTM



LSTM – Long Short Term Memory

Mr. Watson goes on a business trip every Wednesday, and one day during a summer break, her wife was very angry.
Ms. Mary is Mr. Watson's friend, and she is a businesswoman himself.



Memory Cell State :

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Forget Gate :

$$C_t = f_t * C_{t-1}$$

Candidate Value

$$i_t * \tilde{C}_t$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$a_t = \tanh(W_{aa} * a_{t-1} + W_{ax} * x + b_a)$$

$$a_t = \tanh(W_a[a_{t-1}, x_t] + b_a)$$

LSTM – Long Short Term Memory

Mr. Watson goes on a business trip every Wednesday, and one day during a summer break, her wife was very angry. Ms. Mary is Mr. Watson's friend, and she is a businesswoman himself.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

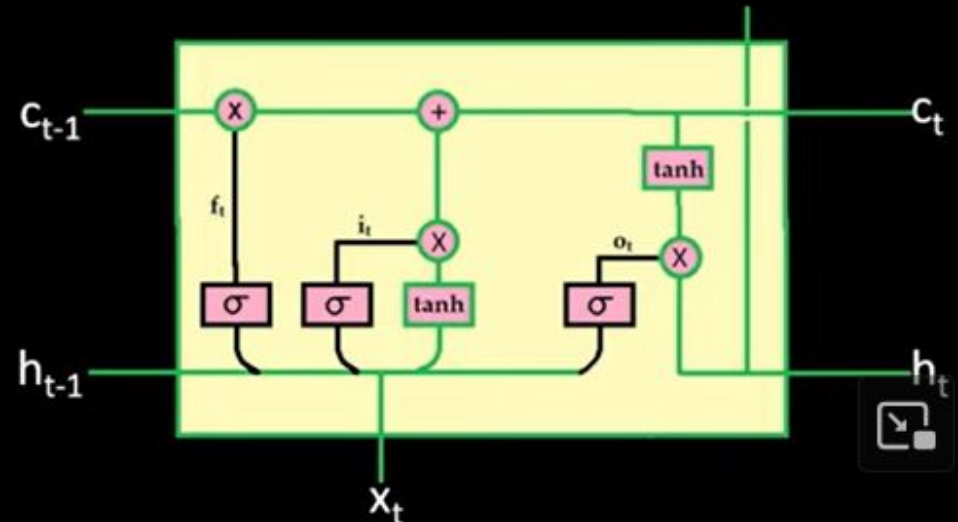
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$



LSTM – Long Short Term Memory

